

Model Paper

Short answer questions

- (1) What are accuracy, precision, recall, and F1-score?
When should each be used?

x Accuracy $\rightarrow$
$$\frac{(TP + TN)}{(TP + TN + FP + FN)}$$

$\rightarrow$ Overall correctness

x Precision $\rightarrow$
$$\frac{TP}{(TP + FP)}$$

$\rightarrow$ When false positives are costly (eg: spam detection)

x Recall $\rightarrow$
$$\frac{TP}{(TP + FN)}$$

$\rightarrow$ When false negatives are costly (eg: medical diagnosis)

x F1-Score $\rightarrow$
$$\frac{2 * (\text{Precision} * \text{Recall})}{(\text{Precision} + \text{Recall})}$$

$\rightarrow$ balanced measure when classes are imbalanced

- (2) Explain the difference between supervised and unsupervised learning.

Supervised	Unsupervised
x Uses labeled data to train models. $\rightarrow$ classification $\rightarrow$ regression	x Finds patterns in unlabeled data $\rightarrow$ clustering $\rightarrow$ dimensionality reduction

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(3.) What is class imbalance in classification, and how can it be addressed?

Occurs when classes are not equally represented

Solutions:

- Resampling (oversampling minority / undersampling majority)
- synthetic data (SMOTE)
- class weights

(4.) What is overfitting in machine learning, and how can it be prevented?

Model performs well on training data but poorly on new data.

prevention:

- Cross-validation
- regularization
- early stopping
- simpler models
- more data

(5.) What is the elbow method in k-means clustering?

Plots WCSS against number of clusters (k)

Optimal k is at the "elbow" where rate of decrease slows

(6.) Mention three ethical considerations in machine learning.

Bias in data / algorithms Data quality issues

Privacy concerns

Bias in training data

Transparency / explainability

Interpretability challenges

(7.) What is classification in machine learning? Give an example.

Process of categorizing data into predefined classes.

ex: - Email spam detection

(8.) Explain the importance of feature selection in supervised learning.

improves model performance

reduces overfitting

faster training

more interpretable models

(9.) How does principal component Analysis (PCA) reduce the dimensionality of data while preserving variance?

transforms data to new orthogonal axes

retains components with highest variance

discards low-variance components

(10.) What is K-means clustering, and how does it work?

partitions data into k clusters

Steps:-

Initialize centroids

assign points to nearest centroid

recalculate centroids

repeat until convergence

(11.) What are some real-world applications of machine learning,

Fraud detection

Recommendation systems

Medical diagnosis

Natural language processing

Image classification

(12.) What is the role of hyperparameter tuning in machine learning?

x Optimizing model parameters not learned during training

x Methods: Grid Search, random search, Bayesian optimization

x Improves model performance.

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(13) How does the choice of k in KNN affect the model's performance? Provide an example.

Small k : More noise-sensitive / complex boundaries

Large k : Smoother boundaries / may miss local patterns

ex:- With $k=1$, a single outlier can misclassify a point

(14) Explain the difference between training and test datasets.

Training $\rightarrow$ used to fit model parameters

Test $\rightarrow$ used to evaluate final model performance on unseen data

Typically split 70-30% or 80-20%

(15) What is the difference between logistic regression and decision trees?

Logistic Regression

Decision Trees

x Linear decision boundary

x Non-linear boundaries

x probabilistic output

x hierarchical splits

x more interpretable

(16) What is elbow method?

- x Technique to determine optimal number of clusters in k -means
- x plots WCSS against k , choose k at "elbow" point

(17) Define dimensionality reduction and explain its importance in machine learning.

- x Reducing number of random variables / features
- x Importance: Curse of dimensionality
 - visualization
 - Computational efficiency

(18) What is the importance of feature scaling in machine learning algorithms?

Distance-based algorithms (KNN, SVM) require scaling

Gradient descent converges faster

Prevents features from dominating due to scale

(19) Discuss one potential issue with k-means clustering and a possible solution.

Problem - Sensitive to initial centroids

Solution - k-means++ initialization

(20) What are the advantages and disadvantages of using k-Nearest (KNN) for classification?

Advantages	Disadvantages
x simple	x computationally expensive
x no training phase	x sensitive to irrelevant features
x works well with small datasets	x curse of dimensionality

Regression Analysis

x	y
Advertising Budget (in \$1000s)	Sales (in \$1000s)
5	52
10	60
15	72
20	78
25	90

- 1.) Identify the independent and dependent variables and fit a simple linear regression model to the data.

(X) independent $\rightarrow$ Advertising Budget

(Y) dependent $\rightarrow$ Sales

Model

$$\hat{y} = \beta_0 + \beta_1 x$$

Sales = $\beta_0 + \beta_1$ (Advertising Budget)

- 2.) Calculate the coefficients β_0 β_1

$$\beta_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

$$= \frac{5 + 10 + 15 + 20 + 25}{5}$$

$$= 15.00$$

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n}$$

$$= \frac{52 + 60 + 72 + 78 + 90}{5}$$

$$= 70.40$$

x	y	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$x_i - \bar{x}^2$
5	52	-10.0	-18.4	184.0	100.0
10	60	-5.0	-10.4	52.0	25.0
15	72	0	1.6	0.0	0.0
20	78	5	7.6	38.0	25.0
25	90	10	19.6	196.0	100.0
				470.0	250.0

$$\beta_1 = \frac{470.0}{250.0}$$

$$\beta_1 = 1.88$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

$$\beta_0 = 70.40 - (1.88 \times 15.00)$$

$$= 70.40 - 28.2$$

$$= 42.20$$

Final Fitted Line

$$\hat{y} = \beta_0 + \beta_1 x$$

$$= 42.20 + 1.88x$$

- 3) Predict the expected sales when the advertising budget is \$18000.

$$x = 18 (\$18000)$$

$$\begin{aligned} y &= 42.20 + 1.88x \\ &= 42.2 + 1.88(18) \\ &= 76.04 \end{aligned}$$

$$76.04 \times 1000 = 76040 \$$$

- 4) Calculate the R^2 value of the model.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

$$R^2 = 1 - \frac{SS_{RES}}{SS_{TOTAL}}$$

x	y	$\hat{y}$	$y_i - \hat{y}_i$	$(y_i - \hat{y}_i)^2$	$y_i - \bar{y}_i$	$(y_i - \bar{y}_i)^2$
5	52	51.60	0.40	0.16	-18.40	338.56
10	60	61.00	-1.00	1.00	-10.40	108.16
15	72	70.40	1.60	2.56	-1.60	2.56
20	78	79.80	-1.80	3.24	7.60	57.76
25	90	89.20	0.80	0.64	19.60	384.16
				7.60	891.20	

$$\hat{y} = 42.2 + 1.88[5] = 51.60$$

$$\hat{y} = 42.2 + 1.88[10] = 61.00$$

$$\hat{y} = 42.2 + 1.88[15] = 70.40$$

$$\hat{y} = 42.2 + 1.88[20] = 79.80$$

$$\hat{y} = 42.2 + 1.88[25] = 89.20$$

$$R^2 = 1 - \frac{7.60}{891.20}$$

$$= 1 - 0.0085$$

$$= 0.9915$$

(x_1)		(x_2)	(y)
Years of Experience	Education Level (1-5)	Salary (Rs. Mn)	
2	3	2.5	
5	4	4.0	
7	5	5.5	
10	5	7.0	
13	3	3.0	

- i) fit a multiple linear regression model to the data.

Step 1 - Construct matrices x and y

$$X = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 5 & 4 \\ 1 & 7 & 5 \\ 1 & 10 & 5 \\ 1 & 13 & 3 \end{bmatrix} \quad Y = \begin{bmatrix} 2.5 \\ 4.0 \\ 5.5 \\ 7.0 \\ 3.0 \end{bmatrix}$$

Step 2 - Compute $x^T x$

First, compute the transpose of x :

$$x^T = \begin{bmatrix} 1 & 2 & 3 & 1 & 1 & 1 \\ 2 & 5 & 7 & 10 & 13 \\ 3 & 4 & 5 & 5 & 3 \end{bmatrix}$$

Now, multiply x^T by x :

$$x^T x = \begin{bmatrix} 5 & 25 & 25 & 25 & 25 \\ 2 & 5 & 7 & 10 & 13 \\ 3 & 4 & 5 & 5 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 \\ 1 & 5 & 4 \\ 1 & 7 & 5 \\ 1 & 10 & 5 \\ 1 & 13 & 3 \end{bmatrix}$$

Perform the matrix multiplication step by step:

$$X^T X = \begin{bmatrix} (1+1+1+1) & (2+5+7+10+13) & (3+4+5+5+3) \\ (2+5+7+10+13) & (4+25+49+100+169) & (6+20+35+50+39) \\ (3+4+5+5+3) & (6+20+35+50+39) & (9+16+25+25+9) \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 5 & 37 & 20 \\ 37 & 347 & 150 \\ 20 & 150 & 84 \end{bmatrix}$$

Step 3 - Compute $X^T Y$

$$X^T Y = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 5 & 7 & 10 & 13 \\ 3 & 4 & 5 & 5 & 3 \end{bmatrix} \cdot \begin{bmatrix} 2.5 \\ 4.0 \\ 5.5 \\ 7.0 \\ 3.0 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} (2.5+4+5.5+7+3) \\ (5+20+38.5+70+39) \\ (7.5+16+27.5+35+9) \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 22 \\ 172.5 \\ 95 \end{bmatrix}$$

Step 4: Find $(X^T X)^{-1}$

$$X^T X = \begin{bmatrix} 5 & 37 & 20 \\ 37 & 347 & 150 \\ 20 & 150 & 84 \end{bmatrix}$$

$$\det(X^T X) = 5 \begin{vmatrix} 347 & 150 \\ 150 & 84 \end{vmatrix} - 37 \begin{vmatrix} 37 & 150 \\ 20 & 84 \end{vmatrix} + 20 \begin{vmatrix} 37 & 347 \\ 20 & 150 \end{vmatrix}$$

$$\begin{aligned}
 \det(X^T X) &= 5(347 \times 84 - 150 \times 150) - 37(37 \times 84 - 20 \times 150) + 20(37 \times 150 - 20 \times 347) \\
 &= 5(29148 - 22500) - 37(3108 - 3000) + 20(5550 - 6940) \\
 &= 5(6648) - 37(108) + 20(-1390) \\
 &= 33240 - 3996 - 27800 \\
 &= 33240 - 31796 \\
 &= \underline{\underline{1444}}
 \end{aligned}$$

* Compute Adjugate (Cofactor Matrix Transpose)

$$X^T X = \begin{bmatrix} 5 & 37 & 20 \\ 37 & 347 & 150 \\ 20 & 150 & 84 \end{bmatrix} \quad \begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

$$\text{Cof}(X^T X) = \begin{bmatrix} (347 \times 84 - 150 \times 150) & -(37 \times 84 - 20 \times 150) & (37 \times 150 - 347 \times 20) \\ -(37 \times 84 - 150 \times 20) & (5 \times 84 - 20 \times 20) & -(5 \times 150 - 20 \times 37) \\ (37 \times 150 - 347 \times 20) & -(5 \times 150 - 37 \times 20) & (5 \times 347 - 37 \times 37) \end{bmatrix}$$

$$\text{Cof}(X^T X) = \begin{bmatrix} (29148 - 22500) & -(3108 - 3000) & (5550 - 6940) \\ -(3108 - 3000) & (420 - 400) & -(750 - 740) \\ (5550 - 6940) & -(750 - 740) & (1735 - 1369) \end{bmatrix}$$

$$\text{Cof}(X^T X) = \begin{bmatrix} 6648 & -108 & -1390 \\ -108 & 20 & -10 \\ -1390 & -10 & 366 \end{bmatrix}$$

* Compute the Adjugate Matrix by Transposing the Cofactor Matrix

$$\text{Adj}(X^T X) = \begin{bmatrix} 6648 & -108 & -1390 \\ -108 & 20 & -10 \\ -1390 & -10 & 366 \end{bmatrix}^T$$

By transposing:

$$\text{Adj}(X^T X) = \begin{bmatrix} 6648 & -108 & -1390 \\ -108 & 20 & -10 \\ -1390 & -10 & 366 \end{bmatrix}$$

Since the cofactor matrix is already symmetric, its transpose is the same.

* Compute the Inverse of $X^T X$

$$(X^T X)^{-1} = \frac{1}{\det(X^T X)} \times \text{Adj}(X^T X)$$

$$\det(X^T X) = 1444$$

$$\text{Adj}(X^T X) = \begin{bmatrix} 6648 & -108 & -1390 \\ -108 & 20 & -10 \\ -1390 & -10 & 366 \end{bmatrix}$$

$$(X^T X)^{-1} = \frac{1}{1444} \times \begin{bmatrix} 6648 & -108 & -1390 \\ -108 & 20 & -10 \\ -1390 & -10 & 366 \end{bmatrix}$$

$$(X^T X)^{-1} = \begin{bmatrix} \frac{6648}{1444} & \frac{-108}{1444} & \frac{-1390}{1444} \\ \frac{-108}{1444} & \frac{20}{1444} & \frac{-10}{1444} \\ \frac{-1390}{1444} & \frac{-10}{1444} & \frac{366}{1444} \end{bmatrix}$$

$$(X^T X)^{-1} = \begin{bmatrix} 4.60 & -0.07 & -0.96 \\ -0.07 & 0.01 & -0.006 \\ -0.96 & -0.006 & 0.25 \end{bmatrix}$$

Step 5: Compute $\beta = (X^T X)^{-1} X^T Y$

$$(X^T X)^{-1} = \begin{bmatrix} 4.60 & -0.07 & -0.96 \\ -0.07 & 0.01 & -0.006 \\ -0.96 & -0.006 & 0.25 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 22 \\ 172.5 \\ 95 \end{bmatrix}$$

$$\beta = \begin{bmatrix} 4.60 & -0.07 & -0.96 \\ -0.07 & 0.01 & -0.006 \\ -0.96 & -0.006 & 0.25 \end{bmatrix} \cdot \begin{bmatrix} 22 \\ 172.5 \\ 95 \end{bmatrix}$$

$$\beta = \begin{bmatrix} (4.60 \times 22) + (-0.07 \times 172.5) + (-0.96 \times 95) \\ (-0.07 \times 22) + (0.01 \times 172.5) + (-0.006 \times 95) \\ (-0.96 \times 22) + (-0.006 \times 172.5) + (0.25 \times 95) \end{bmatrix}$$

$$\beta = \begin{bmatrix} 101.2 - 12.075 - 91.2 \\ -1.54 + 1.725 - 0.57 \\ -21.12 - 1.035 + 23.75 \end{bmatrix}$$

$$\beta = \begin{bmatrix} -2.075 \\ -0.385 \\ 1.595 \end{bmatrix}$$

$$\beta_0 \approx -2.08$$

$$\beta_1 \approx -0.39$$

$$\beta_2 \approx 1.60$$

$$\hat{y} = -2.08 - 0.39 X_1 + 1.60 X_2$$

Model - Regression Analysis

(1.) Final Regression Equation

$$\hat{Y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

$$\text{Salary} = -2.08 - 0.39 X_1 + 1.60 X_2$$

$$\text{Salary} = -2.08 - 0.39 (\text{experience}) + 1.60 (\text{Ed level})$$

(2.)

X_1	X_2	Y
2	3	2.5
5	4	4.0
7	5	5.5

$$1 \longrightarrow 2.5 = \beta_0 + 2\beta_1 + 3\beta_2$$

$$2 \longrightarrow 4.0 = \beta_0 + 5\beta_1 + 4\beta_2$$

$$3 \longrightarrow 5.5 = \beta_0 + 7\beta_1 + 5\beta_2$$

Solve for coefficients

x Subtract (1) from (2):

$$\begin{aligned} 4.0 - 2.5 &= (5-2)B_1 + (4-3)B_2 \\ 1.5 &= 3B_1 + B_2 \rightarrow (A) \end{aligned}$$

x Subtract (2) from (3):

$$\begin{aligned} 5.5 - 4.0 &= (7-5)B_1 + (5-4)B_2 \\ 1.5 &= 2B_1 + B_2 \rightarrow (B) \end{aligned}$$

x Now subtract (B) from (A):

$$\begin{aligned} (3B_1 + B_2) - (2B_1 + B_2) &= 1.5 - 1.5 \\ B_1 &= 0 \end{aligned}$$

Substitute $B_1 = 0$ into (B):

$$\begin{aligned} 2B_1 + B_2 &= 1.5 \\ 2 \times 0 + B_2 &= 1.5 \\ B_2 &= 1.5 \end{aligned}$$

Now use equation (1):

$$\begin{aligned} 2.5 &= B_0 + 2B_1 + 3B_2 \\ 2.5 &= B_0 + 2 \times 0 + 3 \times 1.5 \\ 2.5 &= B_0 + 4.5 \\ -4.5 + 2.5 &= B_0 \\ -2.0 &= B_0 \end{aligned}$$

$$\hat{Y} = -2.0 + 0X_1 + 1.5X_2$$

$$\hat{Y} = -2.0 + 1.5X_2$$

(3.)

$$\hat{y} = -2.0 + 0x_1 + 1.5x_2$$

$$\hat{y} = -2.0 + 0 \times (6) + 1.5(4)$$

$$\hat{y} = -2.0 + 6.0$$

$$\hat{y} = 4.0$$

Predicted salary = Rs. 4.0 Million

Classification

Actual	Predicted	Result
No	No	TN
No	No	TN
Yes	Yes	TP
Yes	No	FN
Yes	Yes	TP
No	Yes	FP
Yes	Yes	TP
No	No	TN
Yes	Yes	TP
Yes	No	FN

TP	—	4
TN	—	3
FP	—	1
FN	—	2

$$(1) a) \text{ Accuracy} = \frac{TN + TP}{TN + TP + FN + FP}$$

$$= \frac{3 + 4}{3 + 4 + 2 + 1} = \frac{7}{10} = 0.7$$

$$= 70\%$$

$$\begin{aligned}
 \text{b) Precision} &= \frac{TP}{TP + FP} \\
 &= \frac{4}{4 + 1} \\
 &= \frac{4}{5} \\
 &= 0.8 \\
 &= \underline{\underline{80\%}}
 \end{aligned}$$

$$\begin{aligned}
 \text{c) Recall} &= \frac{TP}{TP + FN} \\
 &= \frac{4}{4 + 2} \\
 &= \frac{4}{6} \\
 &= 0.667 \\
 &= \underline{\underline{66.7\%}}
 \end{aligned}$$

$$\begin{aligned}
 \text{d) F-measure} &= 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \\
 &= 2 \times \frac{0.8 \times 0.667}{0.8 + 0.667} \\
 &= 2 \times \frac{0.5336}{1.467} \\
 &= 0.727 \\
 &= 0.73 \\
 &= \underline{\underline{73\%}}
 \end{aligned}$$

Yes	-	6
No	-	4
Total	-	10

Initial Entropy

$$E(S) = -P(\text{Yes}) \log_2(P(\text{Yes})) - P(\text{No}) \log_2(P(\text{No}))$$

$$E(S) = -\left(\frac{6}{10} \log_2 \frac{6}{10}\right) - \left(\frac{4}{10} \log_2 \frac{4}{10}\right)$$

$$= -(0.6 \times -0.737) - (0.4 \times -1.322)$$

$$= 0.4422 + 0.5288$$

$$= \underline{\underline{0.971 \text{ bits}}}$$

$$(3.) E(S) = P(\text{Frequent}) * E(1, 3) + P(\text{Moderate}) * E(2, 0) + P(\text{Rare}) * E(2, 2)$$

$$E(S) = \frac{4}{10} * \left(\frac{1}{4} \log_2 \frac{1}{4} \right) - \left(\frac{3}{4} \log_2 \frac{3}{4} \right) + \frac{2}{10} * \left(\frac{2}{2} \log_2 \frac{2}{2} \right) - \left(\frac{0}{0} \log_2 \frac{0}{0} \right)$$

$$+ \frac{4}{10} * \left(\frac{2}{4} \log_2 \frac{2}{4} \right) - \left(\frac{2}{4} \log_2 \frac{2}{4} \right)$$

$$= 0.4 \times [0.25 \times -2] - [0.75 \times -0.415] +$$

$$0.2 \times -[1 \times 0] - [0] + 0.4 \times -[0.5 \times -1] - [0.5 \times -1]$$

$$= 0.4 \times (0.5 + 0.31125) + 0.2(0) + 0.4 \times (0.5 + 0.5)$$

$$= (0.4 \times 0.81125) + (0.4 \times 1)$$

$$= 0.324 + 0.4$$

$$= \underline{\underline{0.724}}$$

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$$IG \text{ (Frequency)} = E(s) - E \text{ after}$$

$$IG = 0.991 - 0.724$$

$$IG = 0.247 \text{ bits}$$

1 → Regression ✓

2 → Clustering ✓

3 → Regression ✓

4 → Classification ✓

5 → Classification ✓

6 → Regression ✓

7 → Classification ✓

8 → Classification

9 → Classification

10 → Classification

11 → Regression ✓

12 → Classification ✓

13 → Classification

14 → Classification

15 → Regression ✓

KNN

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* Consider the following dataset with ten training samples to answer the following questions.

Age	Year of Experience	Monthly Sales	Employee Type
25	3	3000	Junior
45	20	5000	Senior
35	10	4000	Mid-level
50	25	6000	Senior
23	2	2000	Junior
30	5	7000	Mid-level
40	18	12000	Senior
28	4	6000	Junior
33	8	8000	Mid-level
48	22	18000	Senior

Using the above training set, what is the K-nearest neighbor classifier prediction for the Employee Type to the following test example?

(Note: Use Manhattan distance as the distance measure
i.e $d(p, q) = \sum_{i=1}^n |p_i - q_i|$ and K as 3) 3 nearest neighbors

Age	Year of Experience	Monthly Sales (k)
30	5	7000

$$1.) d = (30 - 25) + (5 - 3) + (7000 - 3000) \\ = 5 + 2 + 4000 \\ = \underline{4007}$$

$$2.) d = (30 - 45) + (5 - 20) + (7000 - 5000) \\ = 15 + 15 + 2000 \\ = \underline{2030}$$

$$3.) d = (30 - 35) + (5 - 10) + (7000 - 4000) \\ = 5 + 5 + 3000 \\ = \underline{3010}$$

$$4.) d = (30 - 50) + (5 - 25) + (7000 - 6000) \\ = 20 + 20 + 1000 \\ = \underline{1040}$$

$$5.) d = (30 - 23) + (5 - 2) + (7000 - 2000) \\ = 7 + 3 + 5000 \\ = \underline{5010}$$

$$6.) d = (30 - 30) + (5 - 5) + (7000 - 7000) \\ = 0 + 0 + 0 \\ = \underline{0}$$

$$7.) d = (30 - 40) + (5 - 18) + (7000 - 12000) \\ = 10 + 13 + 5000 \\ = \underline{5023}$$

$$8.) d = (30 - 28) + (5 - 4) + (7000 - 6000) \\ = 2 + 1 + 1000 \\ = \underline{1003}$$

$$\begin{aligned}
 9.) \quad d &= (30 - 33) + (5 - 8) + (7000 - 8000) \\
 &= 3 + 3 + 1000 \\
 &= \underline{\underline{1006}}
 \end{aligned}$$

$$\begin{aligned}
 10.) \quad d &= (30 - 48) + (5 - 22) + (7000 - 18000) \\
 &= 18 + 17 + 11000 \\
 &= \underline{\underline{11035}}
 \end{aligned}$$

Determine the 3 Nearest Neighbors:

Sort the distances and find the three smallest ones:

1. Training Example 6 : Distance = 0 (Mid-level)

2. Training Example 8 : Distance = 1003 (Junior)

3. Training Example 9 : Distance = 1006 (Mid-level)

(1) `df.describe()`

- × Give summary statistics of numerical columns in the DataFrame. (mean, std, min, max)

(2) `df['New-Column'] = df['Old-Column'] * 2`

- × Creates a new column named New-Column by multiplying the values of Old-Column by 2.

(3) `df.fillna(df.mean(), inplace=True)`

- × Fills missing values (NaN) in the DataFrame with the mean value of each column.

- × `inplace=True` means it updates the original DataFrame.

(4) `df.replace({'Male': 0, 'Female': 1}, inplace=True)`

- × Replaces text values - 'Male' becomes 0, and 'Female' becomes 1.
- × Often used to convert categorical data into numbers.

(5) `df.drop_duplicates(inplace=True)`

- × Removes duplicate rows from the DataFrame.

(6) `df[df['Category'] == 'Electronics']`

- × Filters rows where the Category column equals 'Electronics'.

(7) `plt.scatter(df['Age'], df['Salary'])`

- × Creates a scatter plot with Age on the x-axis and Salary on the y-axis using Matplotlib.

(8) `df.duplicated().sum()`

- × Checks for duplicate rows and returns the total number of duplicates.

(9) `df.isnull().sum()`

- × Shows the number of missing values in each column.

(10) `df.hist(figsize=(10,8))`

- x Draws histograms for all numerical columns in the DataFrame
- x `figsize` sets the size of the figure.

(11) `df.drop(['Unwanted-Column'], axis=1, inplace=True)`

- x Deletes a column named 'Unwanted-Column' from the DataFrame
- x `axis=1` → column (use `axis=0` for rows)

(12) `df.to_csv('output.csv', index=False)`

- x Saves the DataFrame to csv file named 'output.csv'
- x `index=False` prevents the row index from being saved.

(13) `x_train, x_test, y_train, y_test, train_test_split(x, y, test_size=0.2, random_state=42)`

- x Splits dataset into training (80%) and testing (20%) sets.
- x Used for machine learning model training / testing.
- x `random_state=42` ensures consistent results each time.

(14) `df = pd.read_csv('data.csv')`

- x Reads a csv file into a Pandas DataFrame.

(15) `df.corr()`

- x Calculates the correlation between all numerical columns.
- x Values range from -1 (negative) to +1 (positive) correlation.

(16) `log_reg = LogisticRegression().fit(x_train, y_train)`

- x Creates and trains a Logistic Regression model using training data.

(17) `sns.heatmap(df.corr(), cmap="coolwarm", annot=True)`

- x Draws a heatmap of the correlation matrix.
- x `cmap="coolwarm"` gives color shades.
- x `annot=True` shows correlation values inside each box.

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Date: ____/____/____

- (18) `tree.plot - tree(decision - tree - model, filled = True)`
- x plots a visual diagram of a trained Decision Tree model.
 - x `filled = True` colors the nodes based on class values.
- (19) `plt.plot (learning - rates, accuracies, marker = 'o')` rate
- x Creates a line plot showing how accuracy changes with learning
 - x `marker = 'o'` adds dots at each data point.
- (20) `sns.lineplot (x = 'Epochs', y = 'Loss', data = loss_df)`
- x Draws a line graph (using Seaborn) showing Loss vs Epochs from the `loss_df` dataset.